AI Tutor: Screenshots of Working Modules

An Evidence-Based Adaptive Learning System with Deliberate Scaffolding Withdrawal

MCA-269 Minor Project, Jagan Institute of Management Studies, Rohini · Guide: Dr. Chetna Laroia

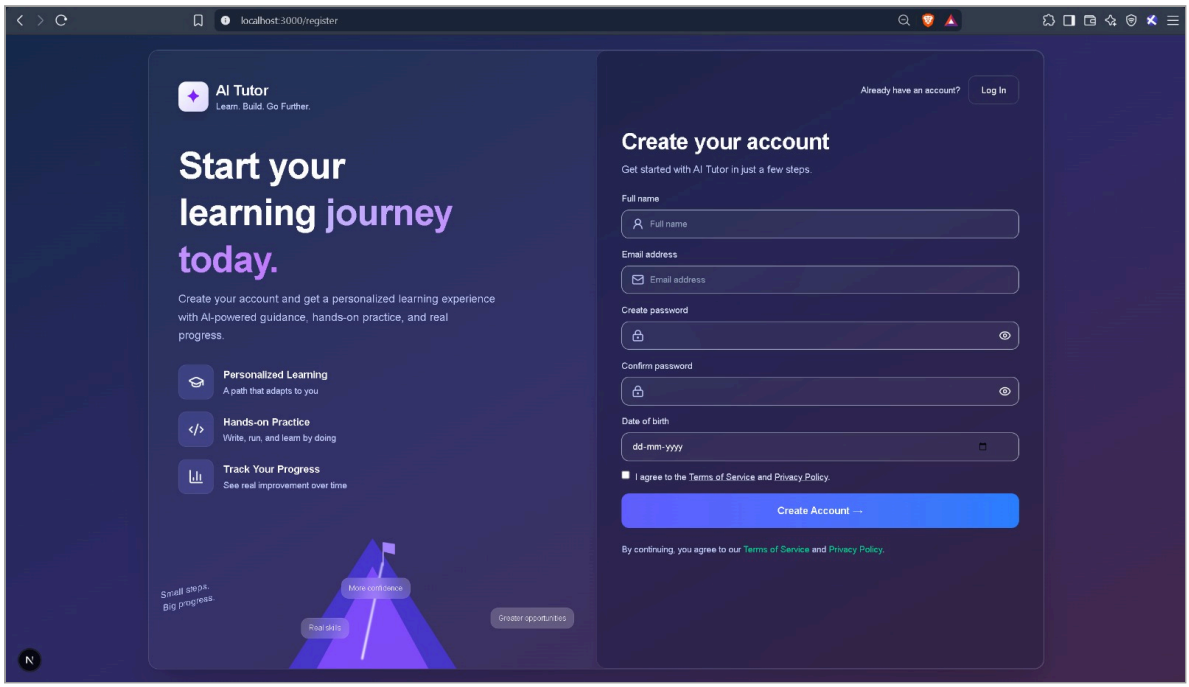
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Taken on 21 October 2026 from the running system: the web app (Next.js) on `localhost:3000`, the API (Express) with PostgreSQL and Redis, and code running in the real E2B sandbox.

No.	Screen	Module
1	Create account	M1 Accounts, consent and data rights
2	Log in, with forgot password	M1 Accounts
3	Privacy policy	M1 Data rights (DPDP Act 2023)
4	The learning loop, step by step	M1, M3–M9, M11, M12
5	Automated test run, part 1	All modules
6	Automated test run, part 2	All modules
7	Automated test run, part 3	All modules
8	Automated test run, part 4	All modules
9	Test results: 235 of 235 passed	All modules

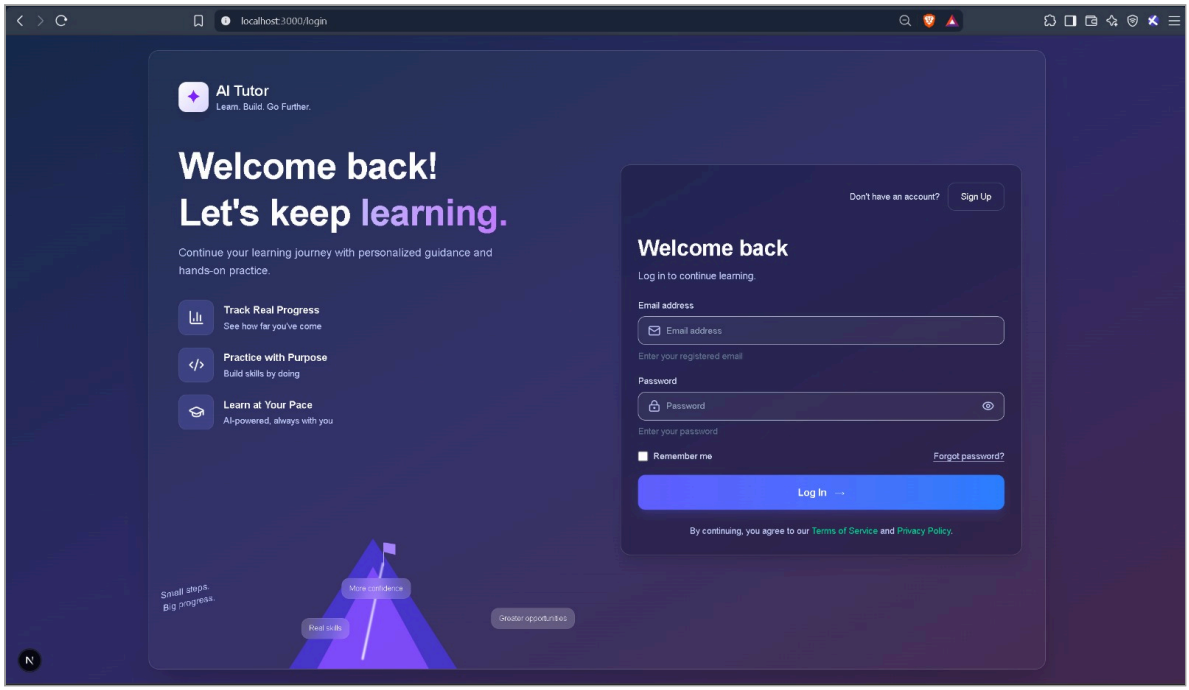
1. Create Account (M1)

Sign-up with full name, email, password, date of birth and agreement to the Terms and Privacy Policy. During the pilot only adults (18+) can register; consent is recorded per purpose with the policy version.



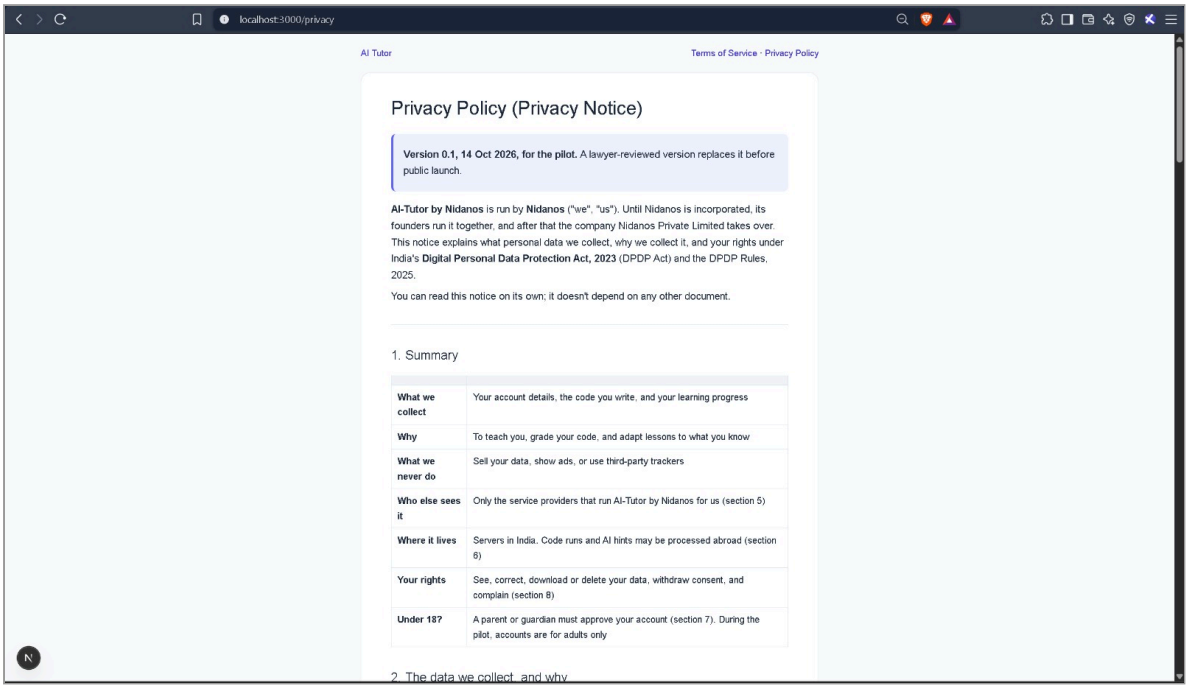
2. Log In (M1)

Login with email and password, "Remember me", and the forgot-password link. A login issues a 15-minute access token and a rotating 30-day refresh token in httpOnly cookies.



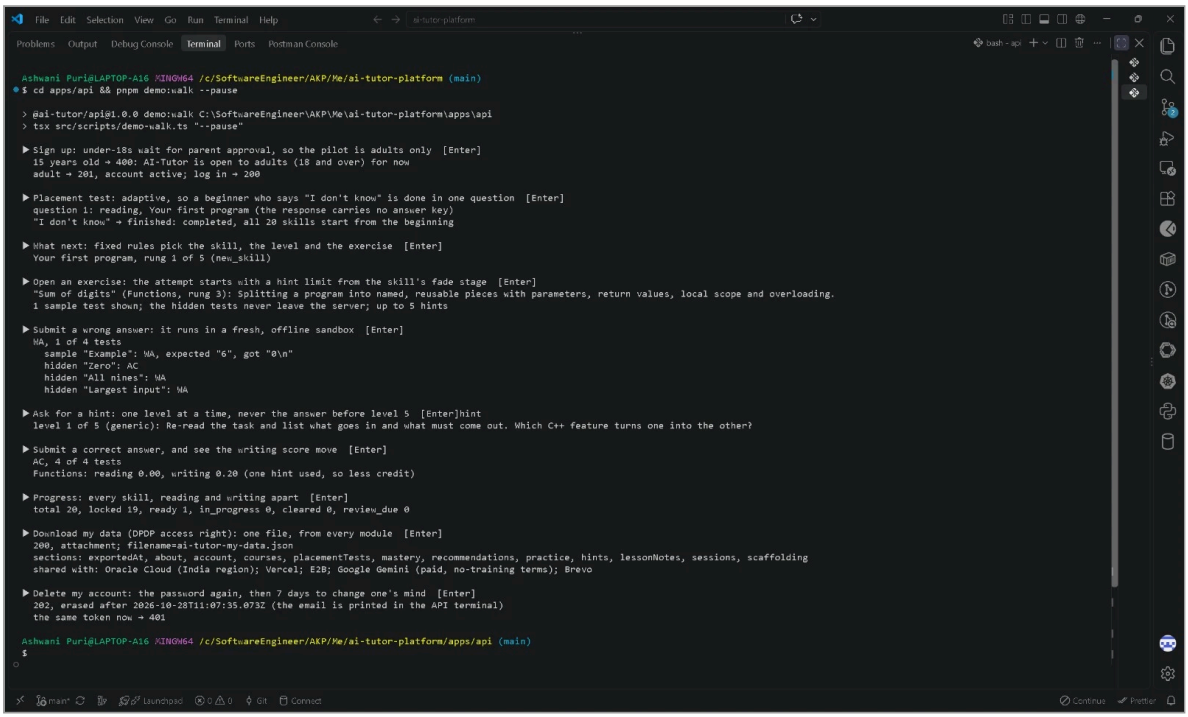
3. Privacy Policy (M1, DPDP Act 2023)

The published privacy notice, built from the same text learners accept at sign-up: what is collected and why, who sees it, where it lives, and the learner's rights.



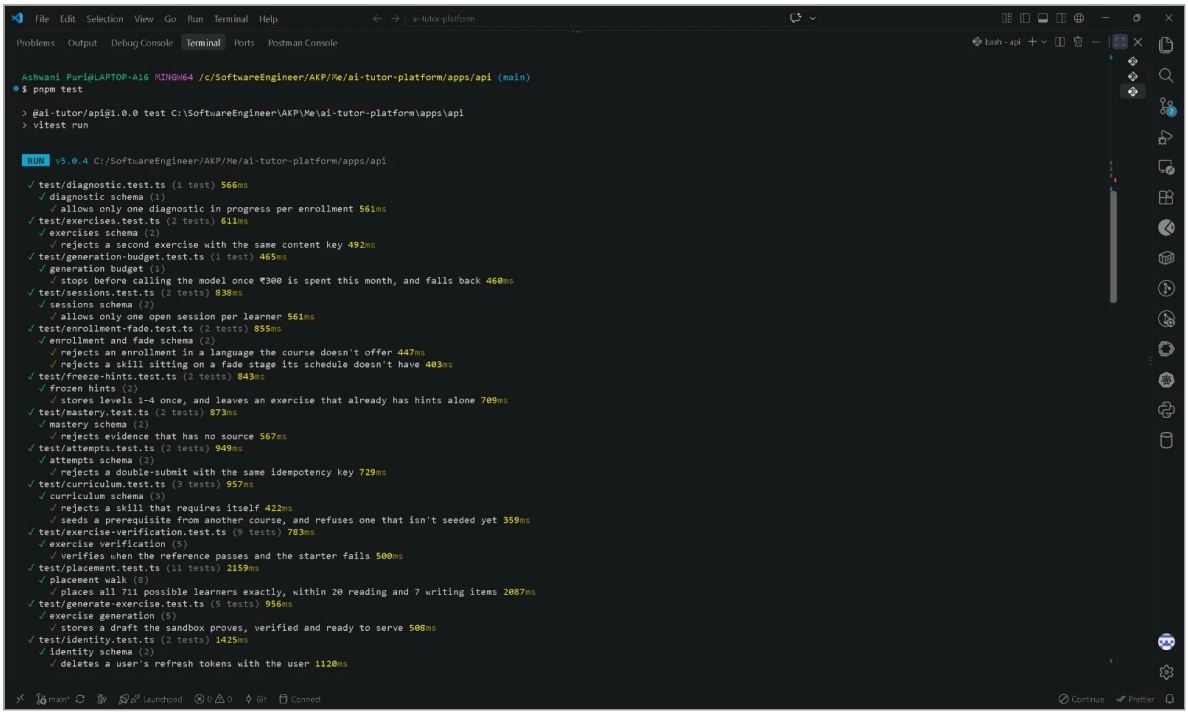
4. The Learning Loop, End to End

A guided walk through the API (`pnpm demo:walk`) on the real code sandbox: adults-only sign-up, the adaptive placement test, the recommender's next step, opening an exercise, a wrong submission graded test by test in a fresh offline sandbox, a hint, a correct submission moving the writing score from 0 to 0.20, progress with reading and writing kept apart, downloading one's data, and deleting the account with a 7-day grace period.



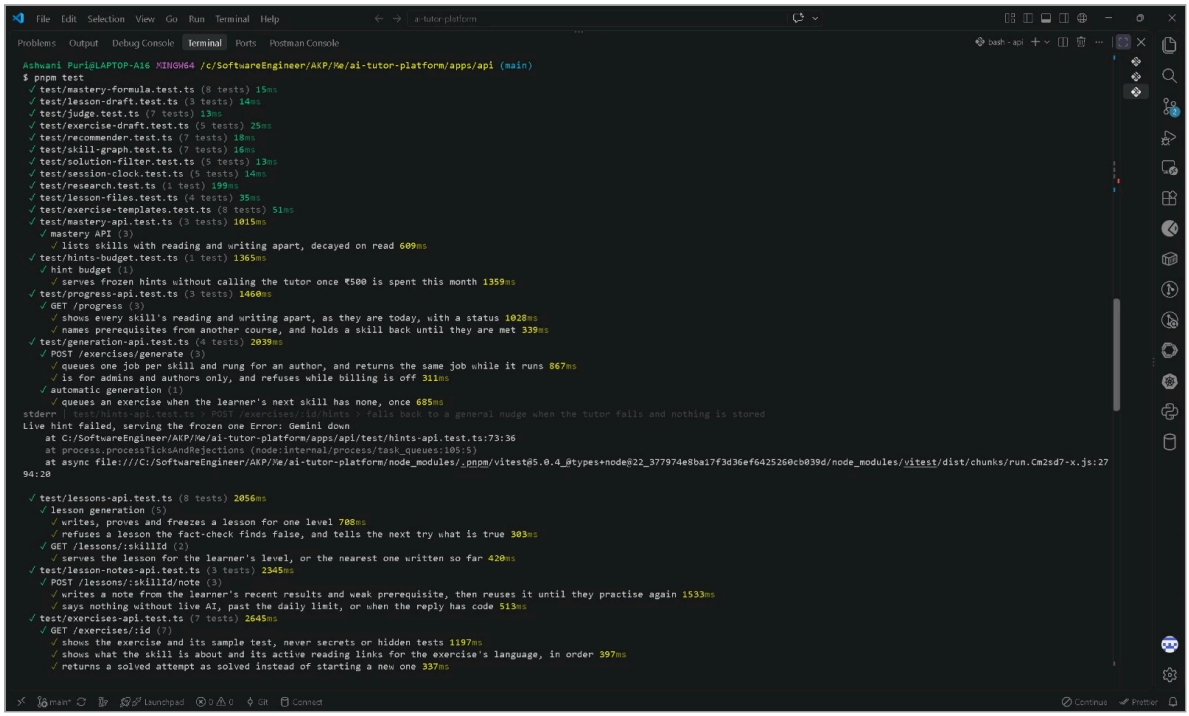
5. Automated Tests, Part 1: Database Rules

Vitest runs the schema tests on a separate test database: one diagnostic per enrolment, unique exercise content, the AI generation budget, one open session per learner, the fade stages, idempotent submissions, the skill graph, exercise verification and the placement walk (all 711 possible learners placed exactly).



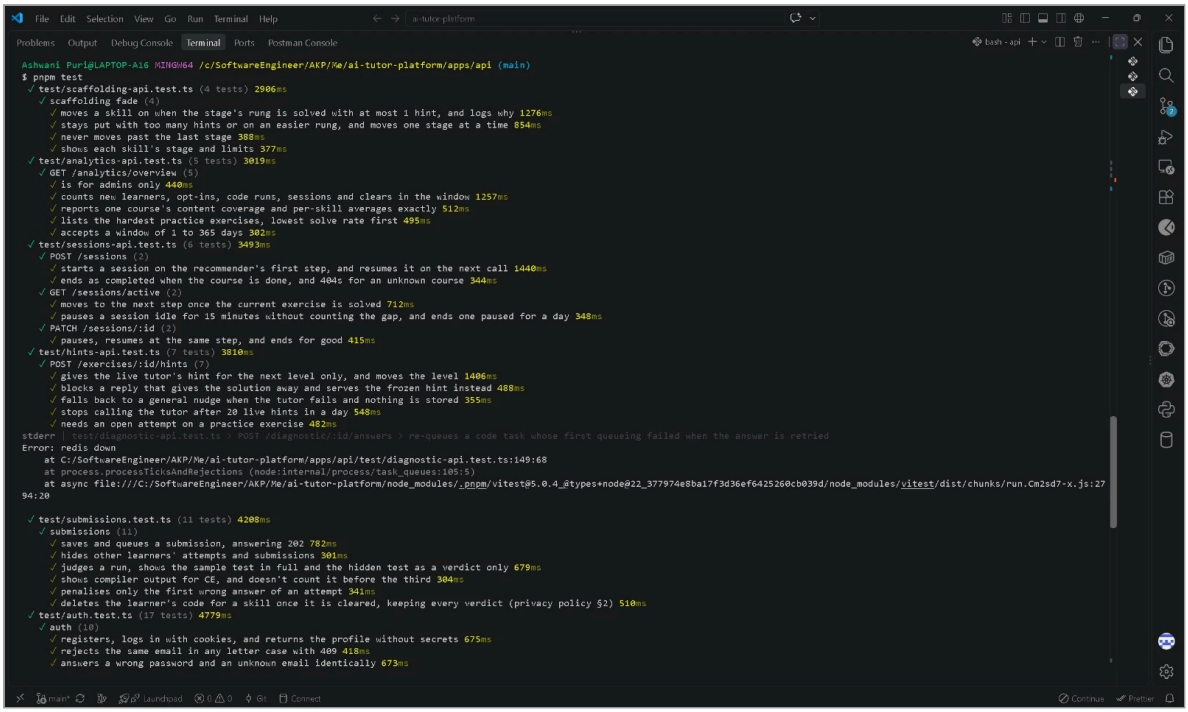
6. Automated Tests, Part 2: Logic and APIs

The mastery formula, the judge, the recommender, the solution leak filter, mastery and progress APIs, exercise generation, lessons and lesson notes. The red "stderr" lines are expected: those tests make Gemini fail on purpose to check the fallback.



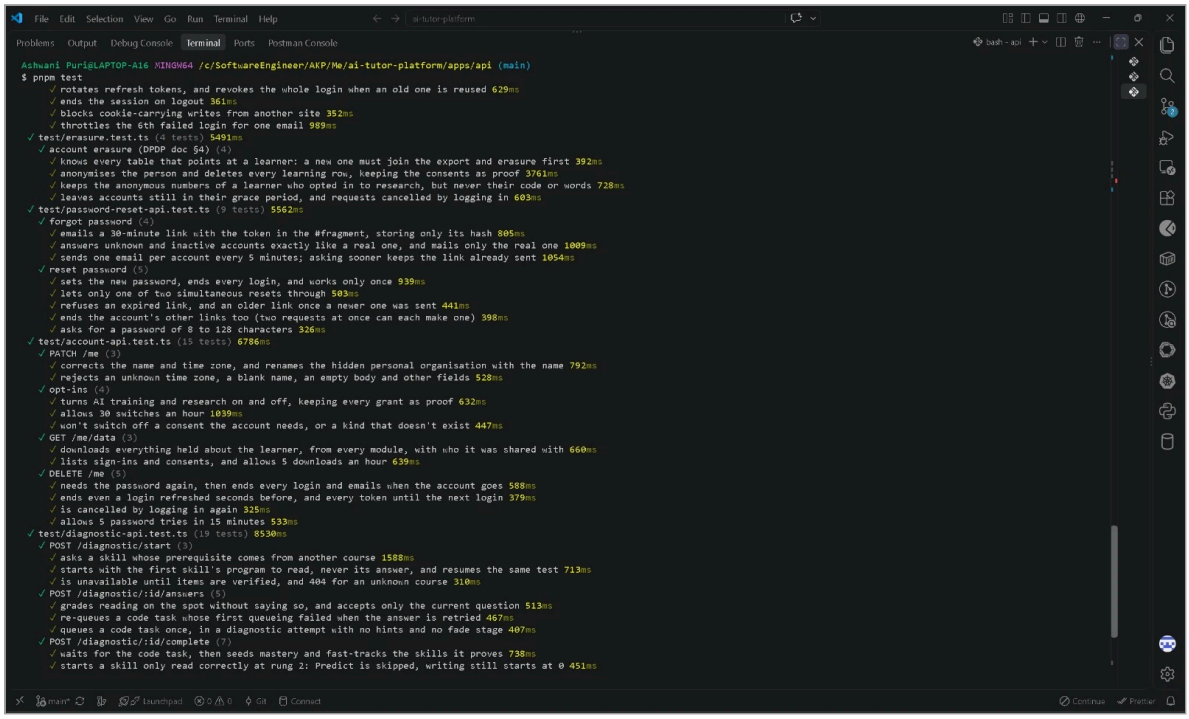
7. Automated Tests, Part 3: Learning APIs

Scaffolding fade, admin analytics, sessions, hints (live tutor, leak filter, fallbacks and daily caps), code submissions and authentication. The "redis down" line is expected: that test makes the queue fail on purposeto check the retry.



8. Automated Tests, Part 4: Accounts, Data Rights and Placement

Token rotation and logout, account erasure (DPDP), password reset, profile correction, consent opt-ins, data download, account deletion, and the placement test API.



9. Test Results

All 42 test files and all 235 tests passed.

